

Business Plan for Decision Tree Classification Assignment

SwiftShip Logistics — Delivery Delay Risk Prediction

Use this business plan together with the Excel dataset provided for the assignment.

1. Business Overview

SwiftShip Logistics is a mid-sized delivery and fulfillment company that provides shipping services for small businesses, retail chains, e-commerce sellers, and enterprise clients across Canada. The company handles shipments from regional warehouses and works with multiple delivery carriers. Its customers expect reliable delivery, accurate communication, and fast action when shipments are at risk.

2. Business Problem

The company is facing a service reliability problem. Some shipments arrive later than promised, especially during high warehouse load, poor weather, long-distance routes, peak seasons, or when carrier performance is weak. Delayed shipments can reduce customer satisfaction, increase support tickets, and damage business relationships.

The business wants to identify shipments that are likely to be delayed before the delivery deadline. This allows operations managers to take preventive action instead of reacting after the customer complains.

3. Machine Learning Decision

Machine learning can help SwiftShip Logistics classify each shipment as either high-risk or low-risk for delay. A Decision Tree classification model is useful because it can produce simple, interpretable rules that managers can understand and discuss.

- If the model predicts a shipment is likely to be delayed, staff may prioritize the shipment, contact the carrier, upgrade the service level, or notify the customer early.
- If the model predicts a shipment is likely to be on time, the company may continue with the normal fulfillment process.
- The prediction supports operational decision-making, but it should not replace human judgment.

4. Target Variable and Input Features

Variable	Type	Meaning
Delivery_Delayed	Binary target	1 = shipment delayed; 0 = shipment delivered on time

The input features describe the shipment, customer, route, warehouse condition, and delivery risk factors.

Feature Group	Examples from Dataset
Customer and business context	Customer_Segment, Region, Product_Category, Order_Value_CAD
Shipping information	Shipping_Method, Shipping_Distance_km, Package_Weight_kg, Promised_Delivery_Days
Operational conditions	Warehouse_Load_Level, Carrier_Rating, Peak_Season
Risk indicators	Weather_Risk, Previous_Delivery_Issues

5. Why This Prediction Is Useful

- It helps the business identify delivery problems before they affect customers.
- It supports proactive customer communication and reduces unexpected service complaints.
- It helps managers allocate resources to high-risk shipments, routes, warehouses, or carriers.
- It can improve customer satisfaction, retention, and operational efficiency.

6. Expected Business Interpretation of Model Results

After training the Decision Tree model, students should explain the model results in business language. The goal is not only to report accuracy, but also to explain what the results mean for SwiftShip Logistics.

Evaluation Concept	Business Meaning
Accuracy	Overall percentage of correct predictions for delayed and on-time shipments.
Precision	When the model predicts a delay, how often that prediction is correct.
Recall	How many truly delayed shipments the model successfully identifies.
F1-score	A balance between precision and recall.
Training vs. testing accuracy	Used to check whether the model may be overfitting.

For this business problem, recall is especially important because missing a shipment that will actually be delayed can lead to poor customer experience, urgent support work, and reputational damage.

7. False Positives and False Negatives

Error Type	Meaning in Dataset	Business Consequence
False Positive	The model predicts the shipment will be delayed, but it is actually on time.	The company may spend extra time or resources on a shipment that did not need intervention.
False Negative	The model predicts the shipment will be on time, but it is actually delayed.	The company may fail to act early, resulting in customer dissatisfaction and service issues.

8. Feature Importance: Business Use

The Decision Tree model may show that some features are more important than others. For example, Warehouse_Load_Level, Weather_Risk, Carrier_Rating, Shipping_Distance_km, Shipping_Method, and Previous_Delivery_Issues may strongly influence whether a shipment is delayed.

- If warehouse load is important, managers may adjust staffing or fulfillment scheduling.
- If carrier rating is important, the company may review carrier contracts or assign reliable carriers to high-value customers.
- If weather risk is important, the company may send early warnings or use alternate routes.
- If previous delivery issues are important, the company may monitor repeated problem routes or customer accounts.

9. Limitations and Responsible AI Reflection

This dataset is a simplified educational dataset. It may not include all real-world factors that influence delivery performance, such as traffic conditions, driver availability, address quality, warehouse system outages, or

sudden carrier disruptions. The model may also reflect past operational patterns, which could create bias against certain regions, routes, or customer groups if the data is not reviewed carefully.

Human judgment is still necessary because operations managers understand context that may not appear in the dataset. A model prediction should be used as decision support, not as the only basis for customer communication, carrier decisions, or resource allocation.

10. Suggested Student Deliverables

- A completed Jupyter Notebook or Google Colab notebook.
- Data preparation and cleaning steps, including missing values and duplicate checks.
- Decision Tree model training and evaluation results.
- Training accuracy and testing accuracy comparison.
- Feature importance chart or table.
- Short business interpretation linked to this business plan.
- GitHub repository link submitted by email or Discord, according to class instructions.